A Partial Map MCMC Algorithm for Addressing Racial Gerrymandering Challenges: A Case Study of Alabama's Congressional Districts

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ABSTRACT

On June 8, 2023, a US Supreme Court ruling preliminarily enjoined Alabama from implementing its recent congressional district plan. The court deemed the plan to be in violation of Section 2 of the Voting Rights Act, which prohibits discriminatory election laws against minority communities. This ruling is related to a century-old problem known as gerrymandering, for which many have sought solutions with the help of mathematics. One key remedy to gerrymandering has been the Markov Chain Monte-Carlo (MCMC) redistricting algorithm, widely employed in partisan gerrymandering litigations to detect outliers in district plans. This study introduces a novel MCMC redistricting algorithm for outlier analysis specific to racial gerrymandering. Our innovative approach, the Partial Map MCMC algorithm, utilizes a subset of districts for plan generation while keeping others static. Testing Alabama's congressional district plan against a standard MCMC algorithm designed for partisan gerrymandering reveals significant differences. Both algorithms were evaluated based on their success in generating district plans with a minimum of two majority-minority districts while maintaining all other constraints constant. Results indicate that the Partial Map MCMC algorithm produces 90% of plans with two majority-minority districts, in stark contrast to the standard MCMC algorithm's less than 1%. This outcome underscores the suitability of our algorithm in generating a diverse set of district plans for outlier analysis in racial gerrymandering litigations. In conclusion, our research represents a crucial step towards addressing the nuanced challenges of gerrymandering, emphasizing the importance of tailored algorithms to promote equitable representation and safeguard the principles of the Voting Rights Act.

CCS CONCEPTS

• **Applied computing** → *Law*; • **Mathematics of computing** → **Graph algorithms**; **Trees**; • **Theory of computation** → **Graph algorithms analysis**; • **Information systems** → **Geographic information systems**.

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KEYWORDS

Redistricting, Gerrymandering, Markov Chain Monte Carlo, Sampling, Simulation, Minority Representation, Racially Fair District Maps, Congressional District Plans

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1 INTRODUCTION

The use of computer algorithms in the generation of election district boundaries has been a subject of extensive research since the 1960s [28]. A substantial body of work has emerged, primarily focusing on combatting partisan gerrymandering [16, 2, 4, 9, 15, 10, 13, 7, 31]. This resulted in the development of various successful algorithms applied in legal disputes related to unfair redistricting [12, 27]. While these efforts have largely concentrated on addressing partisan balance, the impact of unfair redistricting extends beyond political affiliation, encompassing racial disparities. Despite this, the development of algorithms dedicated to the identification and mitigation of racial gerrymandering remains limited [23].

On June 8th, 2023, the United States Supreme Court issued a decisive ruling, preliminarily enjoining the state of Alabama from implementing its recently adopted congressional district plan for the 2022 elections. This landmark decision was prompted by an appeal to the United States District Court for the Northern District of Alabama, which asserted that the adopted plan violated Section 2 of the 1965 Voting Rights Act [1]. The ruling mandates that Alabama swiftly devise a new congressional district map, with a particular emphasis on establishing a second majority-minority district.

A majority-minority district refers to a political district where a minority racial group constitutes the majority of the population within the district. In the case of the Alabama congressional district map, the African American population is considered a minority group in the state. Therefore, any congressional district where African Americans make up the majority of the population is classified as a majority-minority district.

Section 2 of the Voting Rights Act unequivocally prohibits any election laws to discriminate against minority communities, stating that "[t]he right of citizens of the United States to vote shall not be denied or abridged . . . on account of race, color, or previous condition of servitude" [29]. A fair redistricting process must strategically unite groups with significant shared interests to amplify

their influence in government [14]. This approach helps prevent the dilution of voting power in minority communities, thereby upholding the principles outlined in Section 2 of the Voting Rights Act.

To gain a deeper understanding of the Supreme Court ruling, we examine the case of ALLEN, ALABAMA SECRETARY OF STATE, ET AL. v. MILLIGAN ET AL., which prompted the redrawing of Alabama's congressional map. Following the 2020 decennial census, the Alabama legislature's Committee on Reapportionment devised a congressional district map that divided the state into seven congressional districts. Even before the legislature unveiled the new plan, a group of plaintiffs led by Alabama legislator Bobby Singleton filed a lawsuit against the state. They argued that the state's population growth rendered the existing congressional map malapportioned and racially gerrymandered in violation of the Equal Protection Clause [1]. While the litigation was ongoing, the new plan was enacted and subsequently signed into law as HB1. The new map closely resembled the previous decade's map, featuring only one district, district 7 where African American voters constituted a majority. However, some voting rights groups contended that, given that more than a quarter of the state's population is African American, there should be an additional district with an African American majority. The Supreme Court of the United States voted 5-4 in favor of an appeal by voting rights groups, compelling state legislators to devise a new congressional plan [8, 26].

This victory marks a significant milestone in addressing the historical issue of racial gerrymandering, which has perpetuated discrimination against minority populations in the United States. It also raises public awareness of this critical matter and may prompt other states to reevaluate their district maps to ensure compliance with requirements for racial fairness.

This study reviews the current literature on the topic, identifies areas for improvement, and suggests how to use them to reduce unfair redistricting practices. The study is organized as follows: Section 2 provides background information on Alabama's redistricting issue, Section 3 gives a theoretical overview of MCMC redistricting algorithms, Section 4 introduces a new partial map MCMC algorithm designed to create a set of district maps for outlier analysis, Section 5 presents simulations of the algorithm from Section 4, and Section 6 presents the study's discussion and conclusions.

2 PROBLEM OVERVIEW

Before delving into the intricacies of Alabama's congressional map, it is necessary to establish a foundational understanding of the pertinent issue. Gerrymandering, a practice that subverts the fairness of redistricting [17], typically manifests in two principal forms. The first, known as *partisan gerrymandering*, involves the deliberate manipulation of district boundaries to favor one political party or interest group. However, the crux of this research centers on the second form termed *racial gerrymandering*, wherein political boundaries are manipulated to the advantage of a specific racial or ethnic demographic. Racial gerrymandering predominantly culminates in the disenfranchisement and dilution of the political influence of minority racial or ethnic groups.

Irrespective of the category of gerrymandering, two overarching methods are frequently employed to orchestrate the manipulation of district boundaries. These methods are commonly referred to as the *packing* and *cracking* techniques. In the context of racial gerrymandering, both of these stratagems can lead to inequitable and prejudicial representation of minority communities. We briefly explain both techniques:

2.1 Packing

- **Objective**: The goal of packing in racial gerrymandering is to concentrate a significant number of minority voters into a small number of districts.
- **Process**: District boundaries are drawn in a way that includes a large population of minority voters within a limited number of districts. This is done to ensure that these districts have a substantial majority of minority voters.
- **Outcome**: The party in power or those responsible for redistricting may create a few districts with an overwhelming majority of minority voters, allowing minority candidates to win in these districts by large margins.
- **Impact**: While packing may lead to increased representation for minority communities in the districts where they have a majority, it often dilutes their influence in surrounding districts. This can result in fewer overall minority representatives.

2.2 Cracking

- **Objective**: Cracking, on the other hand, aims to disperse minority voters across multiple districts, preventing them from achieving a majority in any one district.
- **Process**: District boundaries are drawn in a way that divides minority communities into various districts, spreading their voting power thinly across those districts.
- **Outcome**: The outcome of cracking is that minority voters are unable to elect candidates of their choice with a majority in any single district, even though they may constitute a significant portion of the total population.
- **Impact**: Cracking diminishes the overall representation of minority communities, making it difficult for them to elect candidates who can effectively advocate for their interests.

Upon an examination of the proposed Alabama congressional map, it becomes evident that a combination of gerrymandering strategies has been employed to limit the presence of African American majorities to just a single district. To illustrate, Table 1a presents the percentages of the three main racial groups within each district.

The population percentages reveal a stark reality among the seven districts: only one has an African American majority. Moreover, this district has an excessive concentration of African American residents, surpassing the white population by a significant 18.9% margin. This clear disproportionality indicates a deliberate effort to pack African American communities into this district.

In a more complex use of gerrymandering tactics, Districts 2, 3, and 7 are neighboring districts, each with a Black population exceeding 20%. The clustering of African American communities in these contiguous districts suggests an intensional strategy to crack these communities, thereby diluting their collective voting power and undermining their ability to exert meaningful influence in the electoral process.

2.3 Race-Neutral Benchmark

In the Supreme Court case Allen, Alabama Secretary of State, et al. v. Milligan et al., Alabama argued that courts should compare the number of majority-minority districts in the challenged map (such as HB1) with the number of such districts in maps created using a race-neutral approach. In these race-neutral maps, race is not considered when drawing district lines. The average number of majority-minority districts in this large set of race-neutral maps would serve as the "benchmark" for determining fairness [1].

Alabama proposed applying the benchmark in the following manner:

(1) **Illustrative Plans:** Plaintiffs challenging the map would need to prove that their example maps are race-neutral, meaning they cannot consider race when designing districts. This approach effectively contradicts the purpose of Section 2 of the Voting Rights Act, which is to address racial discrimination in voting.
(2) **Comparison to the Benchmark:** Courts would then compare the state's enacted map to the average race-neutral map. Alabama argued that if the state's map has a similar number of majority-minority districts as the race-neutral benchmark, it should not be considered in violation of Section 2.
(3) **Prove Race Was the Only Factor:** If plaintiffs demonstrate that the state's map differs from the benchmark, they must further prove that this difference is caused solely by racial discrimination. Alabama argued that other factors, such as geography or population distribution, could not account for the difference—it must be attributed exclusively to race [1].

Alabama's race-neutral benchmark approach would make it significantly harder for plaintiffs to win. By focusing solely on maps drawn without considering race, this method shifts attention away from how racial inequality manifests in real-world voting processes.

The Supreme Court ultimately rejected Alabama's race-neutral benchmark, deeming it inconsistent with the Voting Rights Act's purpose of ensuring minority voters have equal opportunities to participate in elections. The Court emphasized that Section 2 focuses on whether the electoral process is, in practice, fair and open to all voters, not just whether a map appears neutral on its surface.

Experts like Moon Duchin and Kosuke Imai were utilized in this case to provide statistical and algorithmic analyses of Alabama's congressional districting maps. Duchin simulated over two million race-neutral maps, while Imai employed algorithmic techniques to generate districting plans based on traditional principles without considering race. Both experts' analyses showed that most race-neutral simulations produced only one majority-Black district, similar to Alabama's HB1. However, the Court found these algorithmic results insufficient because they failed to account for the broader purpose of Section 2, which addresses the real-world effects of historical and ongoing racial discrimination in voting. While these simulations were technically race-neutral, they could not reflect the realities of racial polarization and social inequality in Alabama. Thus, the experts' findings failed to support Alabama's argument for a "race-neutral benchmark," as Section 2 requires an evaluation of whether minority voters have meaningful opportunities to elect candidates of their choice—something that purely algorithmic models cannot fully measure [1].

The Supreme Court ruling explicitly rejects the argument for a purely race-neutral approach, emphasizing the importance of addressing real-world factors such as racial gerrymandering and ensuring fair representation for minority voters. This decision underscores the need for sophisticated computational methods that can account for these complex realities while adhering to the Voting Rights Act. Consequently, the design of our Partial Map MCMC algorithm, which prioritizes racial distribution as a primary criterion for district plan generation, aligns closely with the Court's ruling. By integrating racial considerations into the redistricting process, the algorithm seeks to create equitable district plans that reflect the practical need for minority voters to have an equal opportunity to elect representatives of their choice, as mandated by the Court's interpretation of Section 2.

3 THEORETICAL BACKGROUND

Among the array of computational redistricting techniques, one of the most effective approaches revolves around the utilization of random walk techniques, such as the Markov Chain Monte Carlo (MCMC) method [11, 12, 20]. These algorithms leverage probabilistic modeling and iterative processes to facilitate the creation of alternative district plans that adhere to predefined criteria and neutral principles.

To further elaborate the theory behind the MCMC redistricting algorithms, we need to consider redistricting as a graph partitioning problem. We set up our initial district plan as a dual graph which will be used as the input to the algorithm.

3.1 Graph (Dual Graph)

The core object in the MCMC redistricting algorithm is the *dual graph* of a geographic partition of the state into units. This graph $G = (V, E)$ includes a vertex for each geographic unit (such as a precinct or block) and edges between units that are geographically adjacent. For example, when applied to Alabama's county map, each county becomes a vertex, and edges are drawn between counties that share borders. To generate a district plan, we partition these vertices into subsets that form connected components, ensuring that each district is geographically contiguous. We assign labels to each vertex representing the district to which it belongs. Additionally, vertices are annotated with data such as population figures, prior election results, and demographic information. This data is used to analyze the plan's attributes, including population balance, partisan balance, and racial balance.

Considering the requirements stated above, we need to use a dual graph instead of a normal graph for the following reasons. In the dual graph, each vertex represents a region, such as a census block, rather than an entity. Each edge represents the adjacency between regions, corresponding to shared boundaries between the faces of the original graph. Furthermore, the graph is derived from a geographical map, which is a planar graph. These factors make a dual graph more suitable for our algorithm than a standard graph.

3.2 State Transisions

A crucial component of the MCMC redistricting algorithm is the method used to determine how to transition from one district plan, or state, to another. To explain further, given a dual graph

$G = (V, E)$, a k-partition of G consists of disjoint subsets $P = \{V_1, V_2, \ldots, V_k\}$ such that $\bigcup V_i = V$. The complete set of k-partitions of G is denoted $P_k(G)$. Each element of $P_k(G)$ is referred to as a districting plan or simply a plan, and each subset V_i is called the ith district. Formally, each X_P is a random variable that describes the transition probabilities between states, with all the values summing to one.

Since $P_k(G)$ is a vast but finite state space, the state transition distribution can be represented as a stochastic matrix. The rows and columns of this matrix are indexed by the states P, with the (P, Q) entry $X_P(Q)$ indicating the probability of transitioning from state P to state Q in a single move. This process forms a Markov chain, where each new state is chosen according to X_P, the distribution based on the current state P [12].

In our study, we utilized a sophisticated redistricting state transition method known as Recombination for the Markov Chain Monte Carlo simulations. This method involves merging two adjacent districts, selecting a *spanning tree* from the merged subgraph, and then cutting it to form two new districts.

In this context, a *spanning tree* is an undirected subgraph that connects all the vertices without forming any cycles. It ensures that all vertices in the merged districts are connected using the minimum number of edges. By removing a single edge from the spanning tree, we can divide the graph into two subgraphs, representing the new districts.

Compared to simpler methods like the Flip transition method, the Recombination method is more computationally intensive. The Flip transition method involves a technique of swapping voters between districts while maintaining population balance.

However, the Recombination method offers superior mixing and distributional properties over the Flip method. These advantages contribute to generating a more diverse ensemble of district plans. The details of the Recombination method and the Flip method are discussed in the study by DeFord (2019) [12].

In addition to the Flip and Recom methods, we have compared our partial map MCMC algorithm with a few other redistricting algorithms. Their theoretical differences are stated below, and some experimental results are posted in Simulations and Findings.

3.2.1 ***Define-Combine Procedure (DCP)*** *[24].*

The DCP algorithm is a method proposed to address partisan gerrymandering by involving both parties in the map-drawing process. It consists of two main stages: *Defining Stage*: One party, referred to as the "Definer," draws a map with double the required number of equal-population contiguous districts. For example, if a state needs to be divided into N districts, the Definer creates a map of 2N subdistricts. *Combining Stage*: The other party, known as the "Combiner," combines these subdistricts into pairs to form the final N districts.

In contrast to our approach, the primary goal of the DCP algorithm is to create a partisan balanced map. Additionally, the DCP method does not employ any nondeterministic approaches, such as the Markov Chain Monte Carlo simulation utilized in our work, nor does it generate an ensemble of district plans for outlier analysis. Consequently, due to the significant differences in objective and approach, it would be challenging to make a reasonable comparison by performing experiments between the two methods.

3.2.2 ***Metropolized Multiscale Forest Recombination (MMFR)*** *[3].*

The Metropolized Multiscale Forest Recombination (MMFR) algorithm improves upon the ReCom algorithm by introducing a hierarchy of quotient graphs, which helps preserve areas of interest [1]from being divided, thus effectively addressing issues such as racial gerrymandering. However, we opted not to utilize MMFR in our study because it might not offer a diverse enough ensemble for outlier analysis. MMFR employs a coarser graph level to prevent the division of areas of interest, resulting in similar shapes for majority-minority districts. In contrast, our algorithm operates at the finest graph level, providing greater diversity in district shapes within majority-minority districts, which is essential for comparing alternative maps.

Furthermore, we conducted experiments using the MMFR algorithm, and the results are available in Simulations and Findings.

3.2.3 ***Sequential Monte Carlo Method (SMC)*** *[21, 22].*

In contrast to both MMFR and our Partial Map MCMC, the Sequential Monte Carlo method (SMC) takes a different approach to the redistricting problem. While MMFR and PM MCMC follow a sampling method that transitions from one district plan to another, the SMC method iterates from district to district by splitting the state into the required number of districts. This method allows for specific constraints, such as compactness and population balance. However, we were unable to impose any specific constraints on this algorithm to avoid the division of areas of interest, leading us to not use it in our experiments.

3.2.4 ***Reversible Recombination (RevReCom)*** *[6].*

The primary difference between the ReCom and RevReCom methods lies in reversibility. In the ReCom method, two adjacent districts are merged and then re-partitioned by cutting a spanning tree, which is not reversible. RevReCom, however, ensures reversibility by carefully constructing a balanced edge cut during the re-partitioning step. This guarantees that the transition between partitions respects detailed balance, making the algorithm reversible and ensuring a proper stationary distribution for unbiased sampling.

We have not identified any significant advancement of this algorithm in our study besides its reversibility. Therefore, we have decided not to include this algorithm in our experiments.

3.3 Map Evaluation Matrices

The following metrics were calculated for each districting plan generated by the algorithms to evaluate three key aspects of fair redistricting practices. To assess racial fairness, the number of Black Majority Districts and the Black Voting Age Population percentages were computed. Compactness was evaluated using the Number of Cut Edges, which measures the geometric integrity of district shapes. Partisan fairness was assessed using the Efficiency Gap, Mean-Median Difference, and Democratic Party Seat Share. Each of these metrics is described in detail below.

3.3.1 ***Number of Black Majority Districts (BMD)***.

The Number of Black Majority Districts metric counts the number of districts

[1]The areas of interest in this context refer to regions with high minority population density, specifically areas with a high density of Black residents. However, A "community of interest," according to Alabama's districting guidelines, is an "area with recognized similarities of interests, including but not limited to ethnic, racial, economic, tribal, social, geographic, or historical identities."

in which Black voters constitute the majority of the Voting Age Population (VAP). This metric is crucial in evaluating racial fairness and compliance with the Voting Rights Act (VRA) [18].

3.3.2 ***Black Voting Age Population Percentage (BVAP).*** The Black Voting Age Population (BVAP) percentage measures the proportion of Black individuals within the Voting Age Population of a district. Higher BVAP percentages in specific districts indicate the ability of Black communities to elect representatives of their choice, as mandated by the VRA.

3.3.3 ***Number of Cut Edges.*** The Number of Cut Edges metric is used to assess the compactness of district shapes. It counts the number of edges in the graph representation of the districting plan that are cut by district boundaries, thus evaluating the geometric contiguity and compactness of the districts [15].

3.3.4 ***Efficiency Gap (EG).*** The Efficiency Gap measures partisan fairness by quantifying the difference in wasted votes between the two major parties across all districts in a plan. A large efficiency gap indicates a plan is biased in favor of one party, with more wasted votes for the disadvantaged party [25].

$$\text{Efficiency Gap} = \frac{W_A - W_B}{V}$$

where:

- W_A is the total number of wasted votes for Party A.
- W_B is the total number of wasted votes for Party B.
- V is the total number of votes.

3.3.5 ***Mean-Median Difference (MMD).*** The Mean-Median Difference metric evaluates the partisan bias of a districting plan by comparing the median vote share with the mean vote share across districts. A large difference suggests a skewed distribution of votes, indicative of potential gerrymandering [30].

$$\text{Mean-Median Difference} = \frac{1}{N}\sum_{i=1}^{N} x_i - \text{Median}(x_1, x_2, \ldots, x_N)$$

where:

- N is the number of districts.
- x_i represents the vote share of the party in district i.
- $\text{Median}(x_1, x_2, \ldots, x_N)$ is the median of the vote shares across all districts.

3.3.6 ***Democratic Party Seat Share (DSS).*** The Democratic Party Seat Share metric calculates the proportion of seats won by the Democratic Party relative to the total number of seats. This measure helps assess whether the seat share aligns proportionally with the overall vote share, reflecting partisan fairness in the allocation of representation [19].

4 PARTIAL MAP MCMC ALGORITHM

As previously discussed, standard MCMC redistricting algorithms are designed to detect partisan gerrymandering in district maps. However, by modifying the graph or the transition method, these algorithms can be adapted to detect racial gerrymandering. To achieve this, it is essential to understand the distinct characteristics of these two types of gerrymandering.

We chose to address racial and partisan gerrymandering as two separate problems due to fundamental differences in their nature. Partisan gerrymandering primarily dilutes votes across the entire state, affecting the overall balance of party representation. Racial gerrymandering, on the other hand, targets specific areas or subsets of districts with high minority population densities, directly impacting the voting power within those communities [23]. Consequently, using MCMC algorithms to tackle both issues simultaneously is not feasible, as each type of gerrymandering requires distinct optimization approaches. This distinction caused us to frame racial and partisan gerrymandering as separate optimization problems rather than a joint optimization of party affiliations and race.

The standard MCMC algorithms possess the capability to stochastically generate a diverse ensemble of district plans. This is achieved through incremental adjustments made to an initial map commonly referred to as the *seed map*. This initial map partitions the geographic region under consideration into a predefined number of districts, each assigned an initial population or voter distribution.

Once the seed map is incorporated into the MCMC algorithm, it undergoes a dynamic transformation process. The algorithm iteratively introduces random modifications to the existing district plan, incorporating operations such as:

- **Swapping voters**: Moving voters from one district to another while maintaining the total population within each district.
- **Splitting or merging districts**: Dividing or combining existing districts to create new ones.
- **Adjusting boundaries**: Adjusting district boundaries slightly to achieve population balance or other criteria.

The distinctive attribute of the Partial Map MCMC algorithm, setting it apart from its conventional counterpart, resides in the seed map. In the context of Partial Map MCMC, the extent of the seed map is intentionally constrained to specific areas of concern within the state map, rather than encompassing the entirety of the state. See Figure 1. This strategic choice is underpinned by two primary objectives:

- **Mitigation of packing**: In scenarios where a particular racial or ethnic group is unjustly concentrated within a single district, the Partial Map MCMC algorithm is deployed to that district and certain adjacent districts. This intervention effectively disperses the densely packed communities among neighboring districts, thereby diminishing the impact of packing.
- **Prevention of further cracking**: By confining the seed map to a localized subset of the state map, the algorithm retains the status quo in the remaining districts throughout the simulation. This strategic decision mitigates any further fragmentation of communities of interest across all districts. Minority communities are contained within designated groups of districts, preventing further dilution of their voting power.

The partial map MCMC algorithm employed in this study was designed based on the Recombination MCMC algorithm. The modified pseudocode for the recom-based Partial Map MCMC algorithm is shown in Algorithm 1.

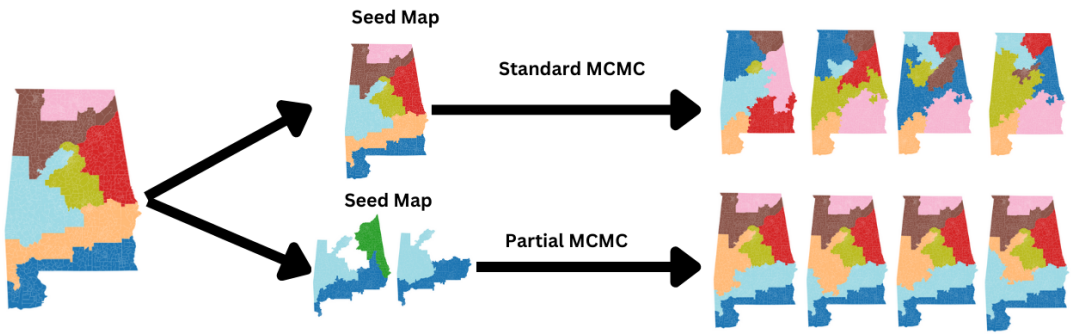


Figure 1: Standard and Partial MCMC Algorithms Seed Maps and Sample Outputs

Algorithm 1: Partial Map MCMC Algorithm

Input: Partial Dual graph $G = (V, E)$, the current partition P, population tolerance ϵ
Output: The next partition Q
Select $(u, v) \in P$ uniformly;
Set $W_1 = P(u)$ and $W_2 = P(v)$;
Form the induced subgraph H of G on the nodes of $W_1 \cup W_2$;
Initialize: *Cuttable* = **False**;
while *Cuttable* = ***False*** **do**
  Sample a spanning tree T of H;
  Let *EdgeList* = [];
  for *each edge edge in T* **do**
    Let $(T_1, T_2) = T \setminus edge$;
    if $|T_1| - |T_2| < \epsilon|T|$ **then**
      Add *edge* to *EdgeList*;
      Cuttable = **True**;
Select *cut* uniformly from *EdgeList*;
Let $R = T \setminus cut$;
Define $Q(v) = \begin{cases} R(v) & \text{if } v \in H \\ P(v) & \text{otherwise} \end{cases}$;
return Q;

As previously stated, the main distinction between the Standard MCMC algorithm and the Partial Map MCMC algorithm lies in the initial configuration (seed map) of the Markov Chain. Hence the dual graph of Algorithm 1 will represent a subset of the nodes and edges from the dual graph of the state. This subset graph will be created by combining the areas with the highest minority populations. To maintain the simplicity of this selection process we suggest picking a subset of adjacent districts with the highest minority population percentages and merging them to form the partial map dual graph.

The partial map dual graph is then used as the initial map for the Markov Chain and forms an MCMC simulation to generate sample district plans. Here we use a spanning tree merge-split algorithm to transition from one district plan to a new district plan. The following steps are taken to transition between district plans.

(1) Randomly select two adjacent districts to merge. The adjacency can be determined based on shared borders or edges in the graph.
(2) Merge the two districts to form a large district. The large district needs to include all the nodes from both districts.
(3) Construct a spanning tree connecting all the nodes in the large district.
(4) Identify a set of edges that can be removed to split the large district into two districts with the population of the two districts nearly equal.
(5) If the population is balanced, split the large district into two by removing the identified edges.

Repeat the process until either (1) the desired number of district plans is achieved or (2) the algorithm finds no possible set of edges to remove to form population-balanced districts.

By incorporating a partial map with the highest minority population into the dual graph, this algorithm limits the dilution of minority voting power into neighboring districts. The algorithm also increases the probability of packing minority communities into districts, leading to increased minority voting power in those districts and an increased number of majority-minority districts.

4.1 Simulation Requirements

The partial map MCMC algorithm can be adapted to apply to any U.S. state with minor modifications to the code and dataset. First, we need to obtain information on the state's census tracts, which is readily accessible through the U.S. Census Bureau's website [5]. This data should include population counts by race within each census tract. Election results data are also needed to evaluate partisan metrics. These two data sources enable the construction of the state's dual graph, which serves as the foundation for running the simulation.

In addition, examining the racial population distribution across the state can guide the selection of districts for the partial map, helping to ensure that the algorithm is targeted effectively. Thus, with basic data preprocessing and minor code adjustments, the algorithm can be modified to fit any other state.

To assess the effectiveness of this new redistricting algorithm, we conducted Partial Map MCMC simulations on select subsets of districts within the state of Alabama. The results of these simulations are presented in the next section. Additionally, for comparative

purposes, we employed the standard MCMC and the Metropolized Multiscale Forest Recombination simulation outcomes as the control state for our experiment, enabling a comprehensive evaluation of the new algorithm's performance relative to the standard redistricting approaches.

5 SIMULATIONS AND FINDINGS

We have categorized our simulations into two groups: Control State Simulations and Partial Map MCMC Simulations. Control State Simulations were carried out to gather data for comparing the effectiveness of our Partial Map MCMC Simulations.

Both simulations utilized two different seed maps of the Alabama congressional district plan. Figure 2 and Figure 3 provide a visual representation of the two distinct initial district plans employed in the simulations:

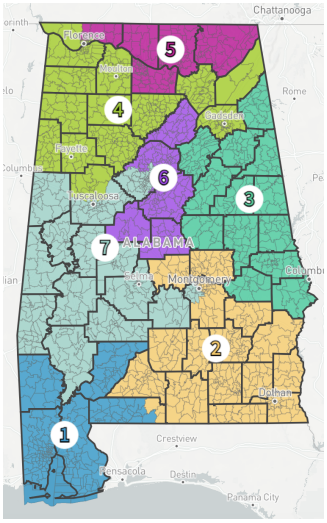


Figure 2: Rejected Alabama Congressional District Plan

- **Figure 2**: This represents the initial 2022 redistricting plan put forth by the legislature. That was later disqualified by the United States Supreme Court. Notably, District 7 emerges as the sole majority-minority district in this plan.
- **Figure 3**: This district plan was generated after the Supreme Court ruling and is currently accepted as the official congressional redistricting plan for the state of Alabama. Notably, there are two majority-minority districts in this plan, Districts 7 and 2.

These distinct initial district plans serve as the starting points for our simulations, enabling a comprehensive evaluation of the proposed algorithm's performance across a spectrum of scenarios and under varying conditions.

To further illustrate the difference in the initial plans, Table 1 shows the population distribution by race in each district.

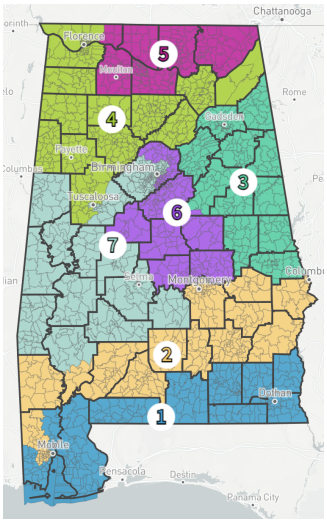


Figure 3: Alabama Congressional District Plan Accepted by Supreme Court

Table 1: Population Percentages By Race in Different Alabama Congressional District Maps.

(a) Rejected Plan

District	White	Black	Hispanic
1	63.4%	25.9%	3.8%
2	59.3%	30.1%	4.4%
3	65.6%	24.8%	3.7%
4	79.9%	7.2%	7.5%
5	68.5%	17.1%	6.6%
6	68.6%	19.0%	6.5%
7	36.5%	55.4%	4.2%
Overall	63.1%	25.6%	5.3%

(b) Accepted Plan

District	White	Black	Hispanic
1	70.5%	18.2%	4.6%
2	43.3%	47%	3.8%
3	69.9%	20.3%	3.8%
4	79.7%	7.5%	7.9%
5	67.8%	17.4%	6.6%
6	71.2%	17.3%	5.5%
7	38.9%	52.3%	4.6%
Overall	63.1%	25.6%	5.3%

5.1 Control State Simulations

In the Control State Simulations, we carried out experiments using two different Markov Chain Monte Carlo (MCMC) redistricting algorithms: the *Recombination* (ReCom) algorithm and the *Metropolized Multiscale Forest Recombination* (MMFR) algorithm. These MCMC simulations were set up to produce a set of 100,000 alternative district plans. Within this simulation, we calculated all the map evaluation matrices mentioned in Section 3 for each individual district across all 100,000 generated plans.

5.2 Partial Map MCMC Simulations

Similar to the Control State Simulations, we conducted multiple simulations using our Partial Map MCMC Algorithm. These simulations were carried out by systematically altering both the initial district plan and the number of districts included in the partial seed map.

The number of districts selected for the partial map can vary depending on population distribution and the anticipated number of majority-minority districts within the state where we run the algorithm. In our study of Alabama's congressional district map, we found that the Black voting population makes up 25.6% of the state. This suggests the potential for creating two majority-minority districts within Alabama's congressional plan. However, selecting four or more districts for the partial map would dilute Black voting power across these four districts, limiting the algorithm's ability to generate plans that achieve two majority-minority districts. Therefore, we decided to limit the number of districts in the partial map to two and three. The details of these simulations are as follows:

5.2.1 Plan A Simulations. Simulations based on Plan A were performed using the plan rejected by the Supreme Court (Figure 2). Two Partial Map simulations were conducted on this seed plan. In the first simulation, we selected the three districts with the highest Black population percentages—Districts 1, 2, and 7—to be included in the partial map. In the second simulation, we focused solely on the two districts with the highest Black population percentages—Districts 2, and 7.

Both simulations employed the Recombination Markov Chain Monte Carlo (MCMC) algorithm. Simulation parameters were set to generate an ensemble of 100,000 distinct district plans.

5.2.2 Plan B Simulations. We adhered to the same structured approach with Plan B, where we also carried out two distinct simulations on the district plan accepted by the Supreme Court (Figure 3). The initial simulation focused exclusively on the three districts characterized by the highest Black population percentages, specifically Districts 2, 3, and 7. This was followed by a simulation with only two districts, 2 and 7.

These simulations retained the same structure as the previous ones, utilizing the recombination algorithm and generating a dataset of 100,000 district plans.

The results of both the control state and partial map simulations are shown in Table 2. The results include average values for each evaluation matrix except the number of Black majority districts.

One of the primary goals of our Partial Map MCMC algorithm is to generate majority-minority districts. Therefore, Figure 4 presents the results showing the number of district plans with 0, 1, and 2 majority-minority districts produced by our algorithm. For comparison, we have also included the number of majority-minority districts generated by the control state simulations.

6 DISCUSSION AND CONCLUSIONS

The preceding section provides a complete analysis of the proposed Partial Map MCMC algorithm with regard to some of the common aspects of computational redistricting algorithms. We will discuss our algorithm's performance in each of these aspects compared to the control state algorithms.

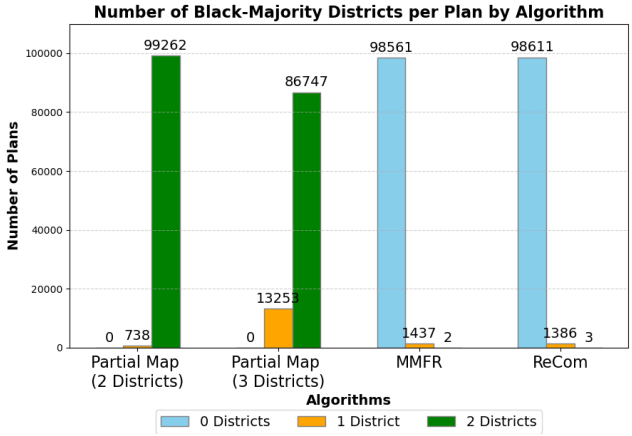


Figure 4: Number of Black Majority Districts Generated from Each Simulation

6.1 Impact on Racial Fairness

As previously stated, the primary objective of our proposed algorithm is to improve the capability of the MCMC redistricting algorithm to address racial gerrymandering issues. Based on the results shown in Figure 4, our Partial Map MCMC algorithm comprehensively outperforms the MMFR and Recom algorithms in generating majority-minority districts. Both the 2-district and the 3-district configurations managed to produce over 80% of maps with two majority-minority districts.

Similar results were observed when we compared the average Black voting age population (BVAP) percentages in each district. Where our algorithm managed to maintain a high BVAP score in majority-minority districts that the MMFR and Recom methods failed to maintain. Figure 5 shows the comparison of BVAP scores per district in Partial Map MCMC vs MMFR and Recom.

6.2 Impact on Compactness

To evaluate the compactness of district shapes, we calculated the number of cut edges in the dual graph. The cut edges count measures the number of edges in the dual graph that are intersected by district boundaries. A high number of cut edges suggests that district boundaries are irregular, indicating lower compactness. As shown in Table 2, the Partial Map algorithm produced a higher cut edge count compared to the Recom and MMFR algorithms, suggesting that the districts generated by our algorithm are less compact. However, the difference is less than 70 edges at most, which remains within an acceptable range. Furthermore, a visual inspection of some of the generated district plans, shown in Figure 6, did not reveal any unusual shapes, indicating that the algorithm still produces reasonable and coherent district maps.

6.3 Impact on Partisan Balance

While the Partial Map MCMC algorithm was not explicitly designed to improve partisan balance, we conducted an evaluation to ensure that it does not introduce any negative effects in this regard. We assessed the algorithm using three partisan balance metrics: Efficiency Gap (EG), Mean-Median Difference, and Democratic Party seat share.

Table 2: Simulations Results for Recom, MMFR, and Partial Map MCMC

Seed Plan	Algorithm	BVAP	Cut Edges	Efficiency Gap	Mean Median Dif	Democratic Party Seat Share
Rejected Map	ReCom	0.26	407.29	0.169607	0.007232	0.49536
	MMFR	0.26	414.28	0.169824	0.007086	0.49438
	Partial Map (3 Districts)	0.26	441.05	0.102541	0.014346	1.00001
	Partial Map (2 Districts)	0.26	430.16	0.099774	0.013878	1.000000
Accepted Map	ReCom	0.26	407.22	0.169033	0.007185	0.49949
	MMFR	0.26	390.58	0.169913	0.007234	0.49373
	Partial Map (3 Districts)	0.26	463.30	0.025492	0.060897	1.98816
	Partial Map (2 Districts)	0.26	472.26	0.034926	0.065348	1.99996

The Efficiency Gap measures the extent of partisan gerrymandering by calculating the difference between the wasted votes of two parties in an election. A higher EG score suggests a greater potential for gerrymandering. As shown in Table 2, the Partial Map MCMC algorithm achieved a comparatively low EG score compared to the Recom and MMFR algorithms, indicating a better partisan balance in the maps generated by our approach.

On the other hand, the Mean-Median Difference shows a slight increase relative to Recom and MMFR, suggesting a minor skew in the distribution of votes and potential for partisan imbalance.

Additionally, we observed a rise in the Democratic Party seat share compared to both the Recom and MMFR algorithms. While this does not necessarily imply partisan bias, it is an outcome worth noting for further analysis.

6.4 Impact on Democratic Party Seat Share

The influence on Democratic Party seat share (DSS) can be assessed in two key ways. In our simulations, we used the 2020 presidential election results to calculate DSS values. While our study does not aim to establish a direct correlation between minority voters and Democratic voters, a comparison of Alabama's Black voting population distribution with the election results map reveals a notable alignment. Specifically, areas with a high Black population density tend to vote predominantly for the Democratic Party. This observation suggests that enhancing Black voting power within specific districts may contribute to an increased Democratic vote share in these districts, but this was not a consideration in the design of the procedure itself.

Secondly, it is important to note that the DSS values in Table 2 are averages across 100,000 maps in each simulation. The rejected and accepted Alabama congressional plans contain one and two Democratic congressional districts, respectively, based on the 2020 presidential election results. Standard MCMC simulations yielded mean DSS values around 0.5, classifying both the rejected and accepted maps as outliers. We would expect to see a mean DSS closer to 1 for the rejected plan and 2 for the accepted plan. Importantly, our partial map MCMC algorithm brings these values closer to those anticipated. In the 2020 presidential election, the Democratic Party received 37% of the overall vote in Alabama, supporting the presence of two Democratic seats in the state. Thus, district plans generated by our algorithm not only improved racial fairness but, as an unintended consequence, enhanced partisan fairness as well.

6.5 Impact on Markov Chain Convergence

The convergence of a Markov Chain refers to the point at which its sampling distribution stabilizes and accurately represents the target distribution. In MCMC redistricting algorithms, it is not strictly necessary to discard samples generated before convergence. However, discarding maps produced during the burn-in period—before convergence is achieved—can enhance the representation of the target distribution. Another way to address this issue is by optimizing the MCMC algorithm to converge more quickly, thereby reducing the burn-in period.

To evaluate the impact of our algorithm on Markov Chain convergence, we used the trace plot method. This involves tracking the values of a specific parameter (e.g., district compactness, partisan bias) over time to observe whether the chain fluctuates around a central value without any discernible trends. Figure 7 shows a comparison of the efficiency gap trace for the Partial Map MCMC algorithm versus the Recom and MMFR algorithms. The results indicate that our algorithm reaches convergence faster, maintaining stability around the central value, which suggests a more efficient and stable sampling process.

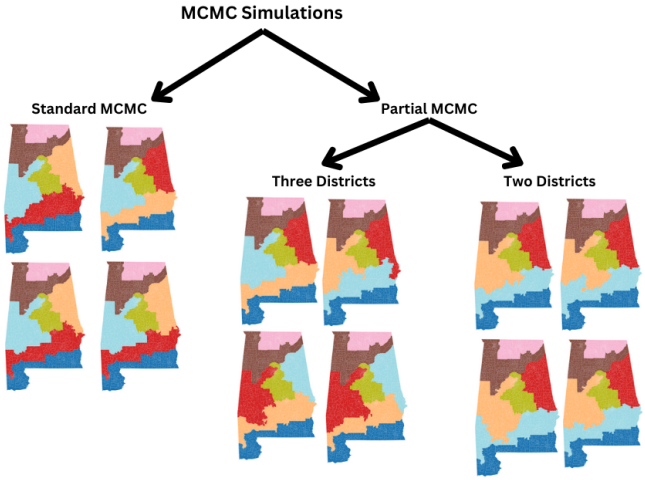
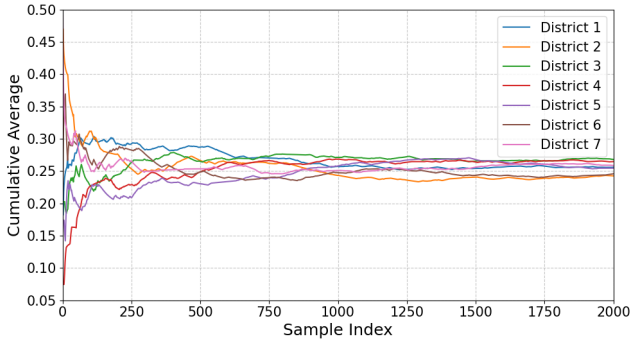
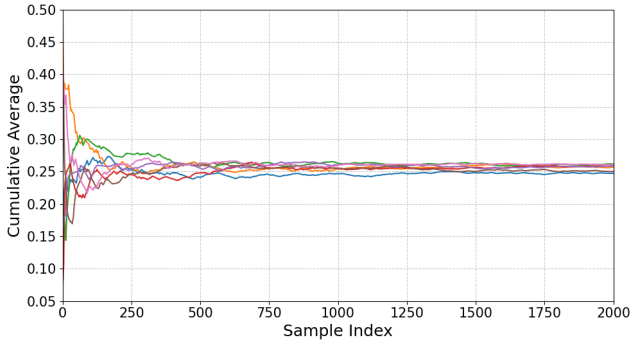


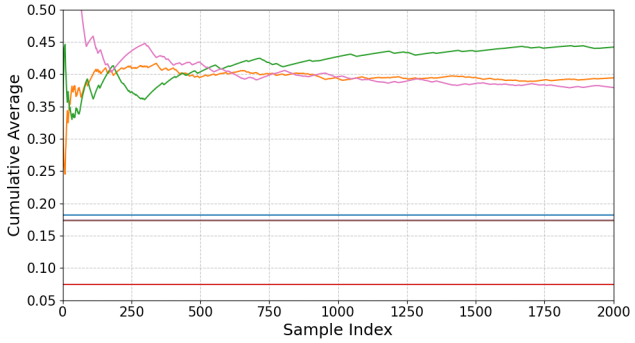
Figure 6: Sample Outputs Generated from Each Simulation



(a) Cumlative BVAP for Recom Algorithm



(b) Cumlative BVAP for MMFR Algorithm



(c) Cumlative BVAP fo Partial Map MCMC Algorithm with 3 Districts

Figure 5: BVAP Percentage Per District Comparison

6.6 Conclusions

The findings of this study provide compelling evidence that the Partial Map MCMC algorithm is significantly more effective in producing racially fair district maps compared to traditional MCMC methods. We compared the performance of the Partial Map MCMC algorithm against the Metropolized Multiscale Forest Recombination and Recombination algorithms. Our method consistently outperformed these conventional algorithms in generating district plans with majority-minority districts.

Furthermore, we evaluated the algorithm's performance with regard to partisan balance, compactness, and convergence. The results indicate that, with the exception of certain compactness metrics,

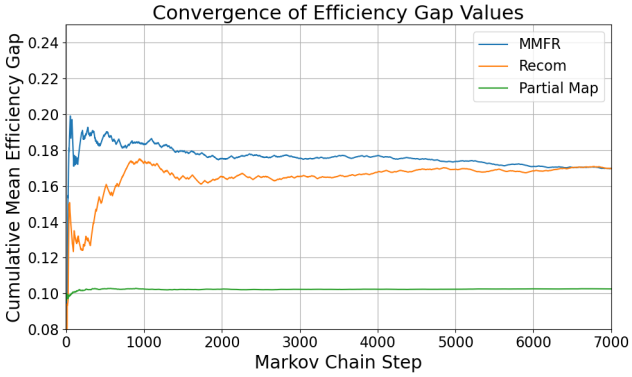


Figure 7: Convergenece of the Efficiency Gap Values

the Partial Map MCMC algorithm exhibited superior performance across all other map evaluation metrics.

This study primarily focused on the Alabama congressional district map. To further substantiate our findings, we conducted preliminary tests using the South Carolina congressional district map. The results, detailed in Appendix A, demonstrate that our Partial Map MCMC algorithm similarly outperforms traditional algorithms in this context.

In summary, the Partial Map MCMC algorithm offers an enhanced and effective approach for generating political district maps that uphold the principles of racial fairness and equitable representation.

7 FUTURE WORK

This study concentrated on Alabama, a state where the standard MCMC algorithm struggles to generate a second majority-minority district, despite the Supreme Court's ruling that one is clearly warranted based on population data. Although we conducted a preliminary test on South Carolina, a comprehensive analysis is necessary to further validate the efficacy of our algorithm. States like Louisiana, which face similar redistricting challenges, present compelling opportunities for testing and refining the Partial Map algorithm.

Moreover, Alabama's unique geographic and demographic configuration, where the two majority-minority districts are contiguous, highlights a specific scenario. Future research should focus on adapting the Partial Map algorithm for states where majority-minority districts are geographically separate and do not share boundaries. This adaptation would expand the applicability of the algorithm and enhance its robustness in various redistricting contexts.

The study presented in this paper primarily addresses the computer science and mathematical aspects of the racial redistricting problem. This focus stems from the authors' limited expertise in legal matters. We recommend that future research delve into the legal implications of racial gerrymandering, particularly examining the impact of algorithmic approaches, such as the Partial Map MCMC algorithm, on addressing this complex issue.

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A APPENDIX A: SOUTH CAROLINA TEST

South Carolina's district map is shown in Figure 8. It has 7 congressional districts, and District 6 is the only majority-minority one.

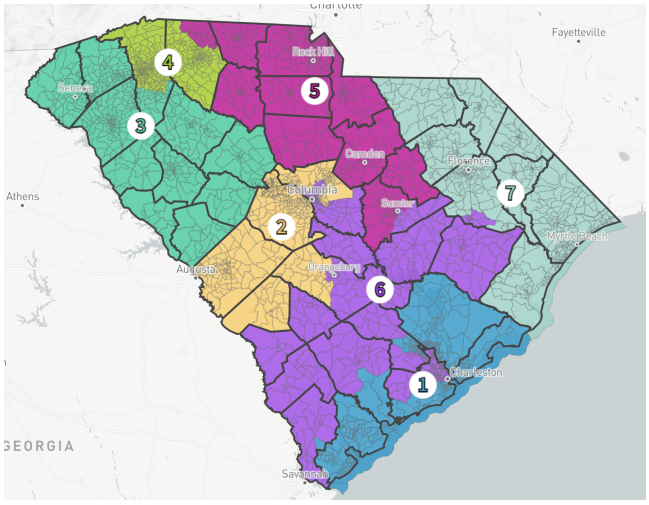

Figure 8: South Carolina Congressional District Map

We conducted simulations using both the partial map MCMC and ReCom algorithms to assess their effectiveness in producing district plans with majority-minority representation. For the partial map MCMC simulation, we used Districts 6 and 7 as the fixed partial map, while the ReCom algorithm was applied to the entire map

of the state. Both simulations were set to generate 100,000 maps, keeping all other parameters consistent across both methods. The final outcome of the simulations is presented in Figure 9.

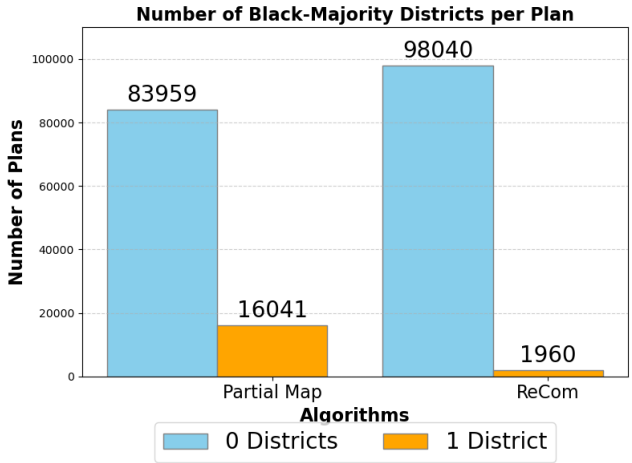

Figure 9: South Carolina Simulation Results

Again, the Partial Map MCMC algorithm outperformed the Standard ReCom algorithm, producing a little over 8 times as many maps with a majority-minority district.